

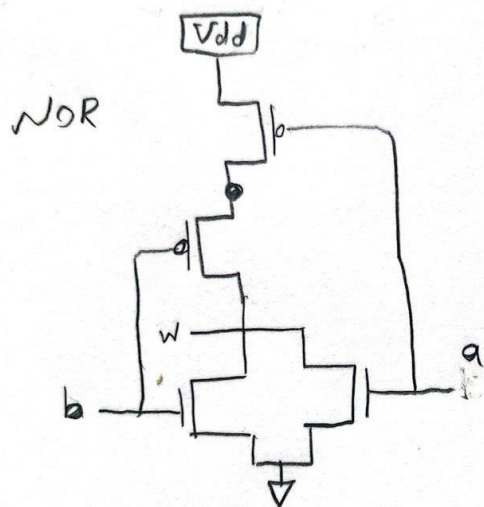
IN THE NAME OF GOD

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SID: 810102601 , LOGIC Number: 92

1)

NOR:



Worst case to 0 →

$$\max(7+7, 4) = 14 \text{ ns}$$

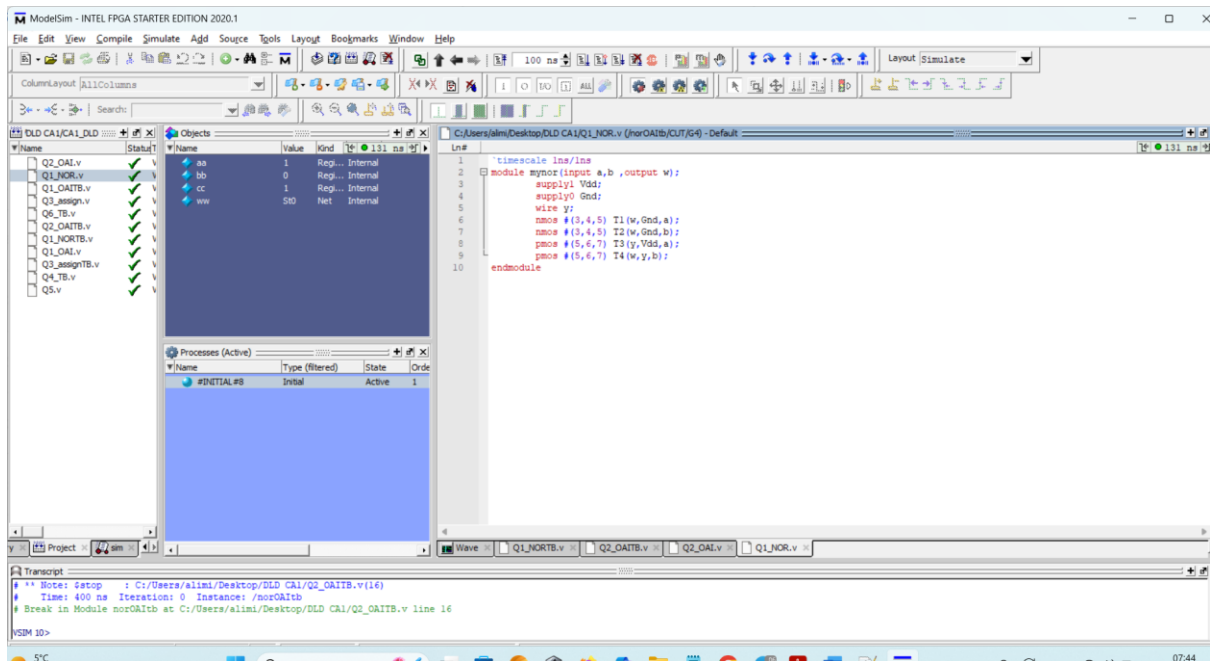
Worst case to 1 →

$$\max(5+5, 5) = 10 \text{ ns}$$

Verilog description

```
*C:\Program Files\Notepad++\new 1.v - Notepad++ [Administrator]
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window 2
new 1.v
1 timescale 1ns/1ns
2 module mynor(input a,b ,output w);
3 supply1 Vdd;
4 supply0 Gnd;
5 wire y;
6 nmos #(3,4,5) T1(w,Gnd,a);
7 nmos #(3,4,5) T2(w,Gnd,b);
8 pmos #(5,6,7) T3(y,Vdd,a);
9 pmos #(5,6,7) T4(w,y,b);
10 endmodule

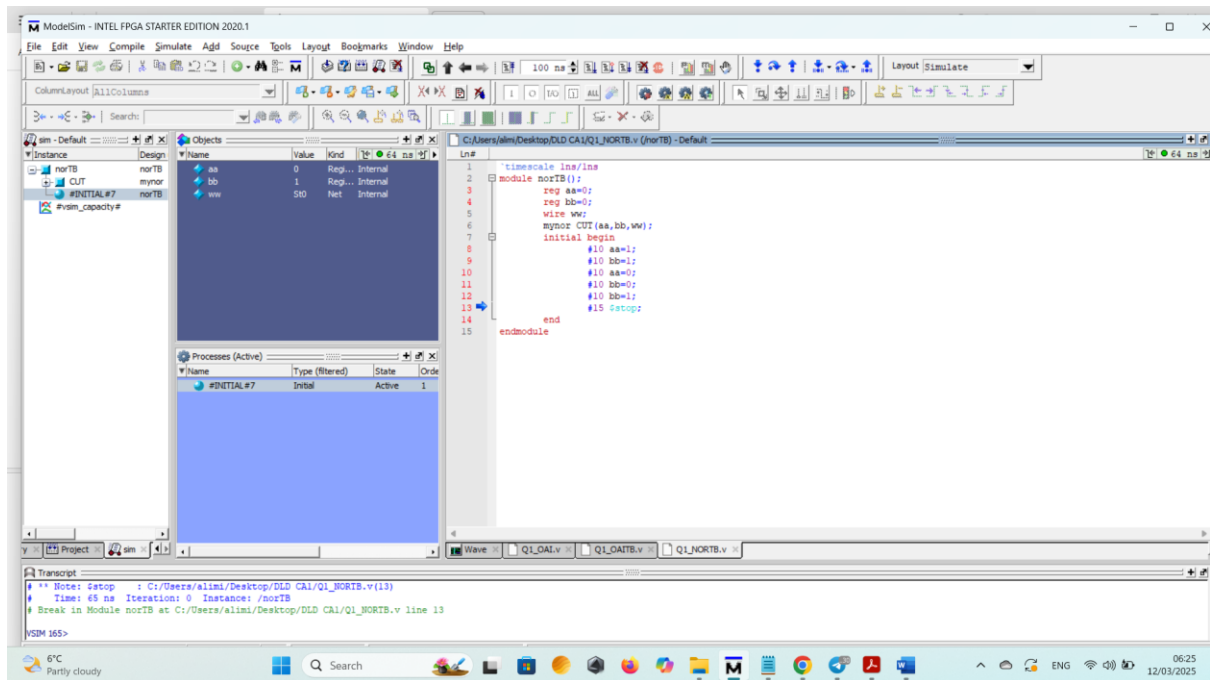
Verilog file length: 219 lines: 10 Ln: 10 Col: 10 Pos: 220 Windows (CR LF) UTF-8 INS
```



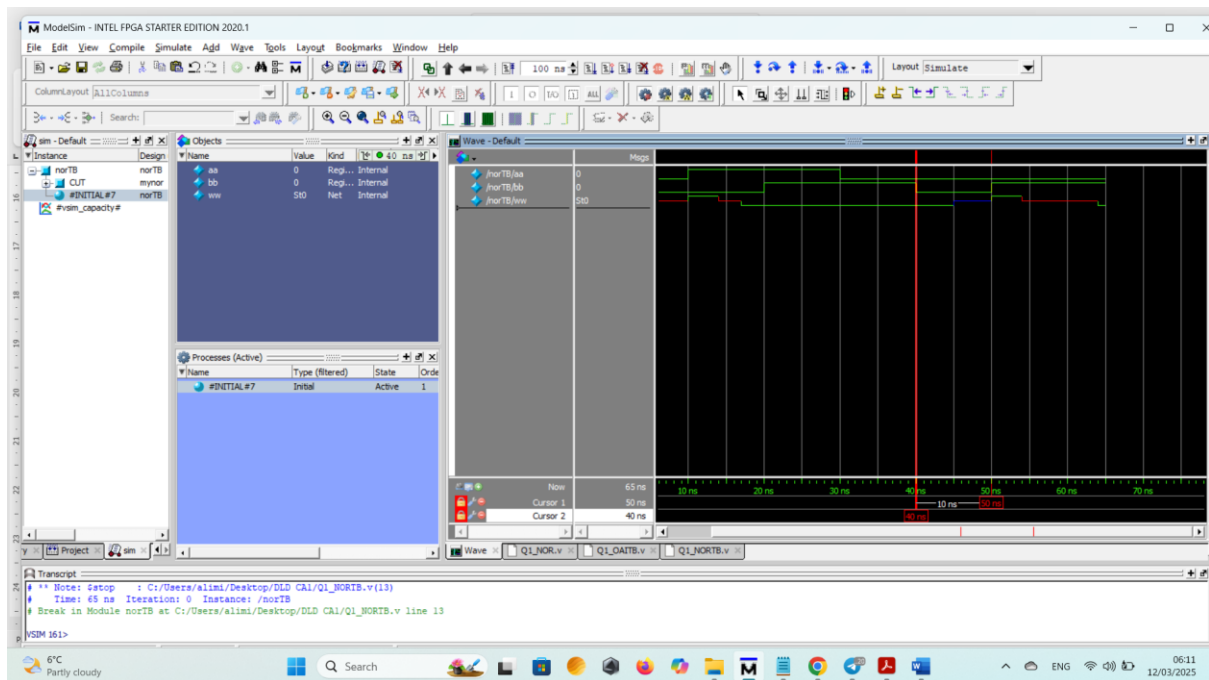
Verilog testbench

```
*C:\Program Files\Notepad++\new 1.v - Notepad++ [Administrator]
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window 2
new 1.v
1 timescale 1ns/1ns
2 module norTB();
3 reg aa=0;
4 reg bb=0;
5 wire ww;
6 mynor CUT(aa,bb,ww);
7 initial begin
8     #10 aa=1;
9     #10 bb=1;
10    #10 aa=0;
11    #10 bb=0;
12    #10 bb=1;
13    #15 $stop;
14 end
15 endmodule
```

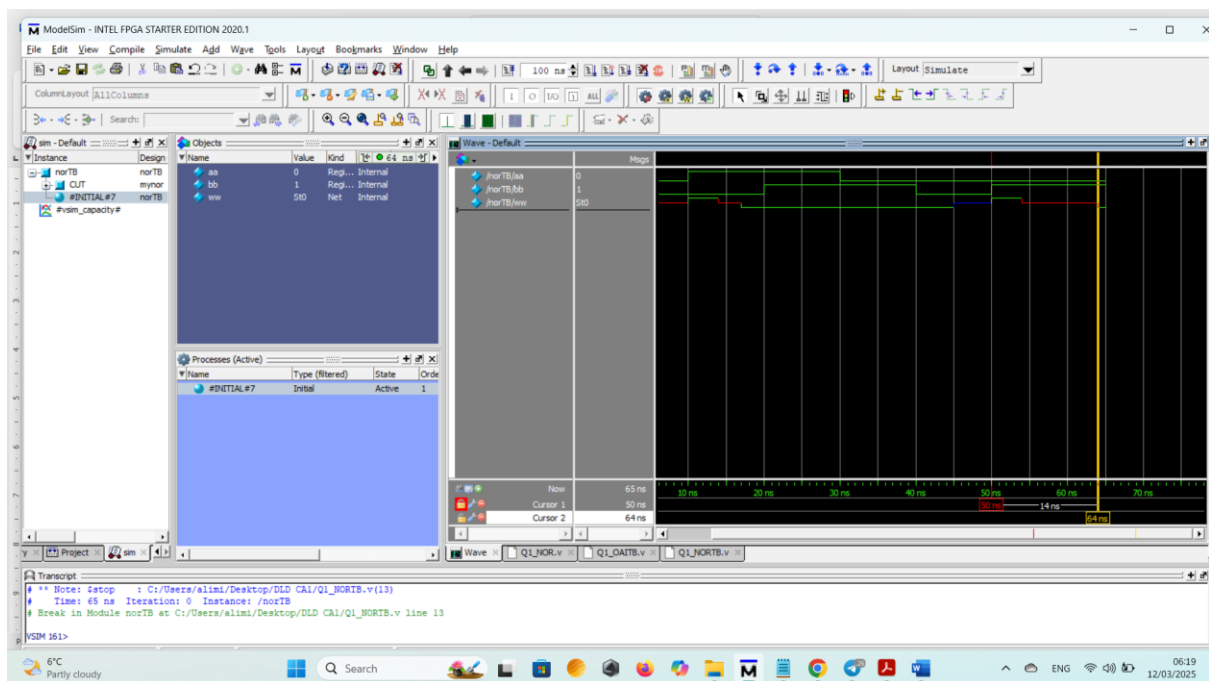
Verilog file length: 205 lines: 15 Ln: 15 Col: 10 Pos: 206 Windows (CR LF) UTF-8 INS



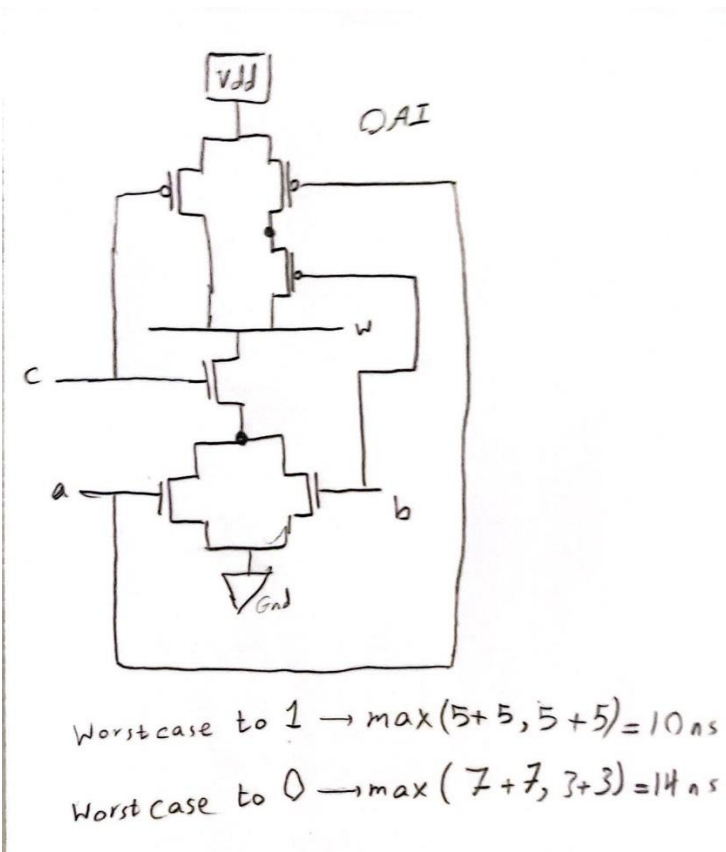
Worst case to 1:



Worst case to 0:



OAI:



Verilog description:

```

1  `timescale 1ns/1ns
2  module myQAI(input a,b,c, output w);
3      wire u; wire v;
4      supply1 Vdd; supply0 Gnd;
5      nmos # (3,4,5) T1(v,Gnd,a), T2(v,Gnd,b), T3(w,v,c);
6      pmos # (5,6,7) T4(u,Vdd,a), T5(w,u,b), T6(w,Vdd,c);
7  endmodule

```

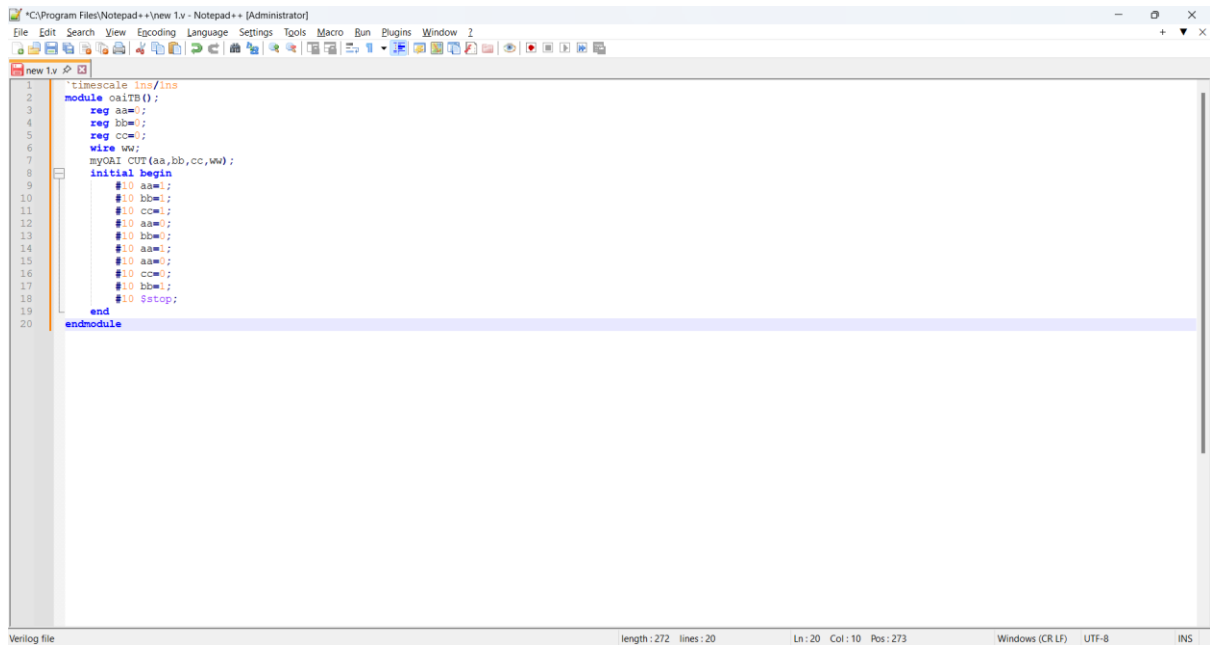
Verilog file length: 219 lines: 7 Ln: 7 Col: 10 Pos: 220 Windows (CR LF) UTF-8 INS

```

1  `timescale 1ns/1ns
2  module myOAI(input a,b,c, output w);
3      wire u; wire v;
4      supply1 Vdd; supply0 Gnd;
5      nmos #(3,4,5) T1(v,Gnd,a), T2(v,Gnd,b), T3(w,v,c);
6      pmos #(5,6,7) T4(u,Vdd,a), T5(w,u,b), T6(w,Vdd,c);
7  endmodule

```

Verilog testbench



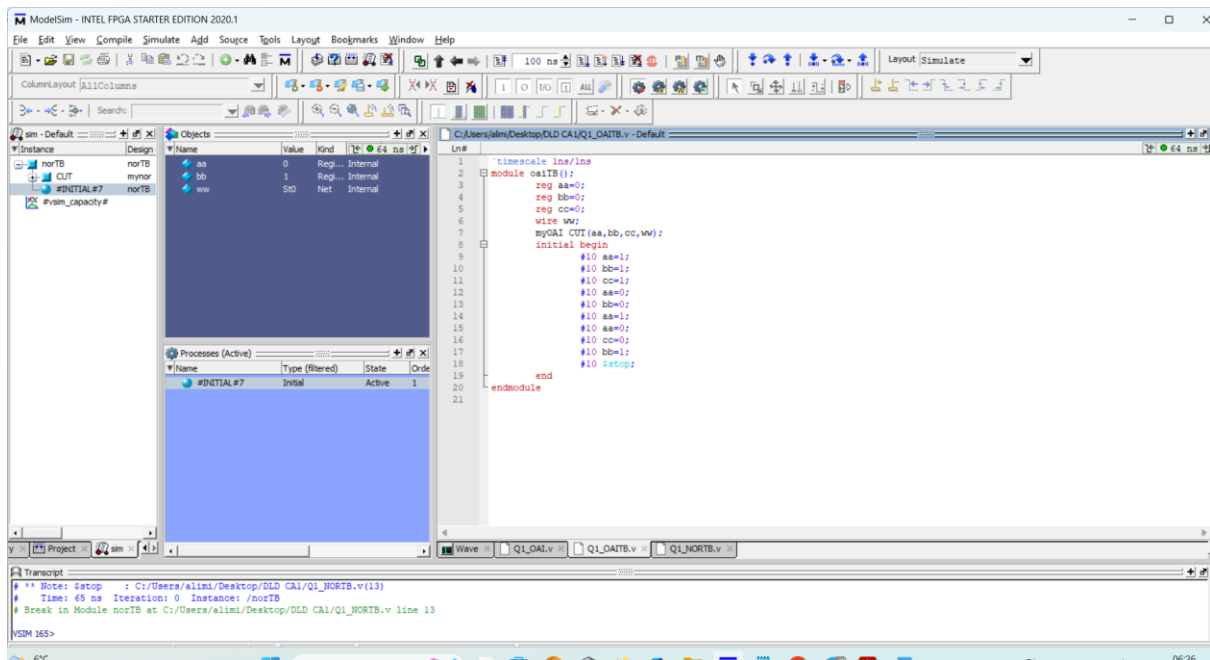
The screenshot shows a Notepad++ window titled "new 1.v" containing a Verilog testbench. The code is as follows:

```

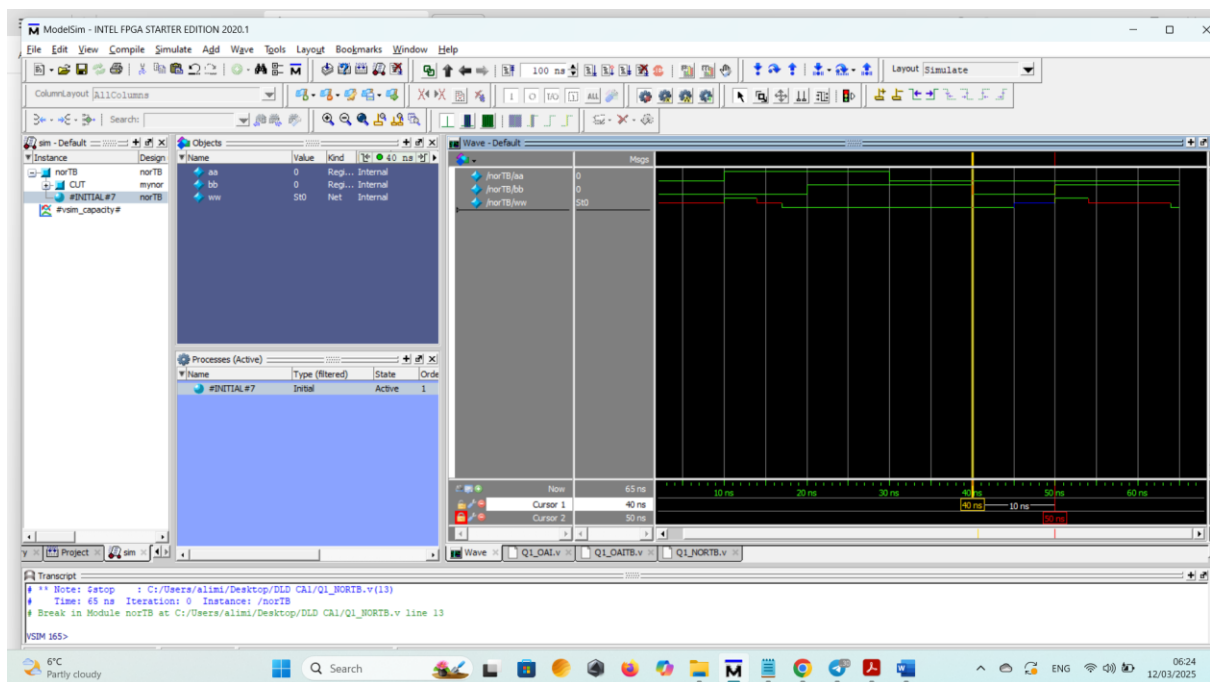
1  `timescale 1ns/1ns
2  module ositB1();
3      reg aa=0;
4      reg bb=0;
5      reg cc=0;
6      wire ww;
7      myOAI CUT(aa,bb,cc,ww);
8      initial begin
9          #10 aa=1;
10         #10 bb=1;
11         #10 cc=1;
12         #10 aa=0;
13         #10 bb=0;
14         #10 aa=1;
15         #10 aa=0;
16         #10 cc=1;
17         #10 bb=1;
18         #10 $stop;
19     end
20 endmodule

```

The status bar at the bottom indicates "Verilog file", "length: 272 lines: 20", "Ln: 20 Col: 10 Pos: 273", "Windows (CR LF)", "UTF-8", and "INS".



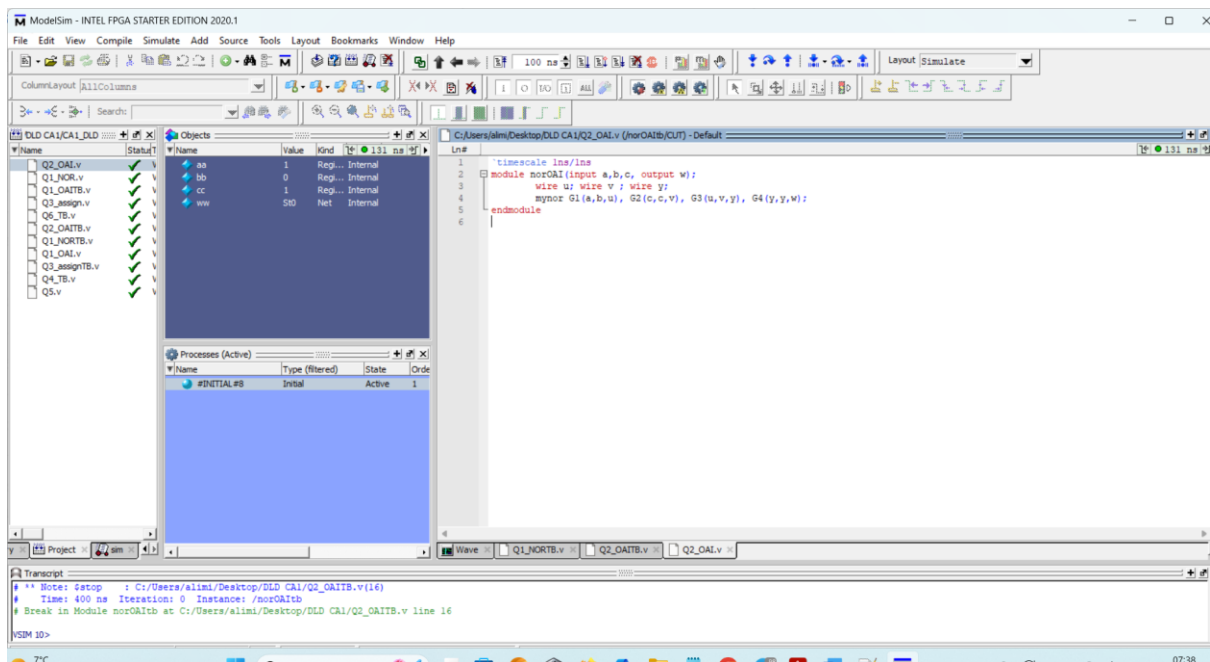
Worst case to 1:



Worst case to 0:


```
*C:\Program Files\Notepad++\new 1.v - Notepad++ [Administrator]
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window 2
new 1.v
1 timescale 1ns/1ns
2 module norOAI(input a,b,c, output w);
3   wire u; wire v; wire y;
4   mynor G1(a,b,u), G2(c,c,v), G3(u,v,y), G4(y,y,w);
5 endmodule
```

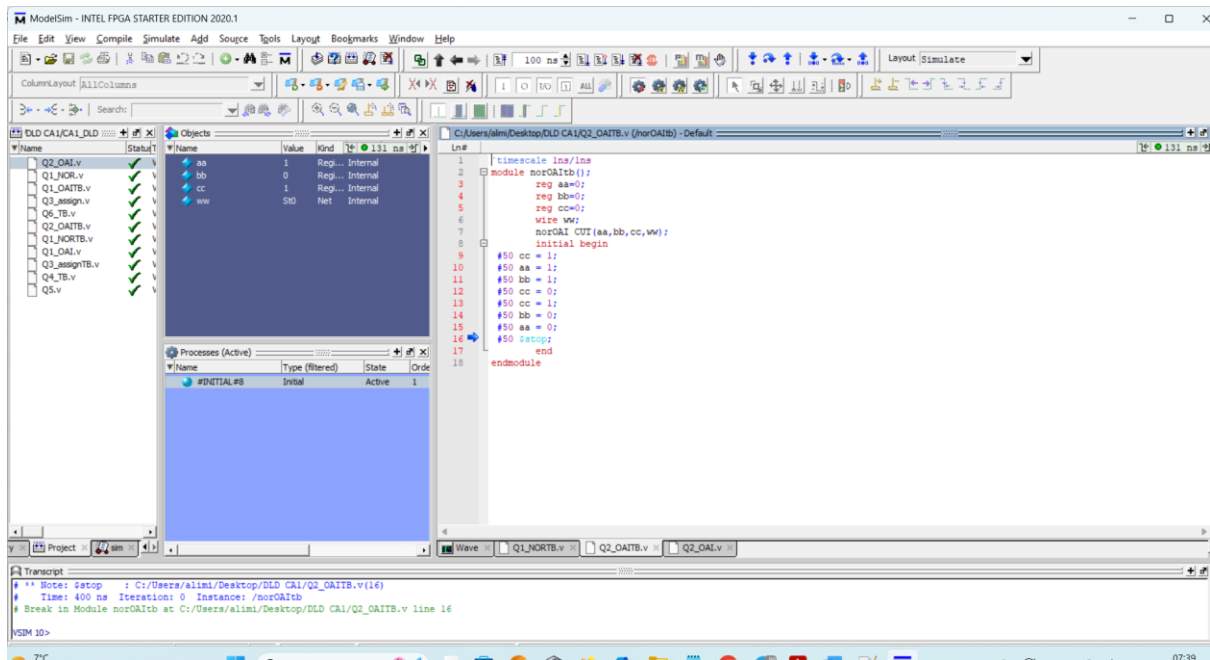
Verilog file length: 147 lines: 5 Ln: 5 Col: 10 Pos: 148 Windows (CR LF) UTF-8 INS



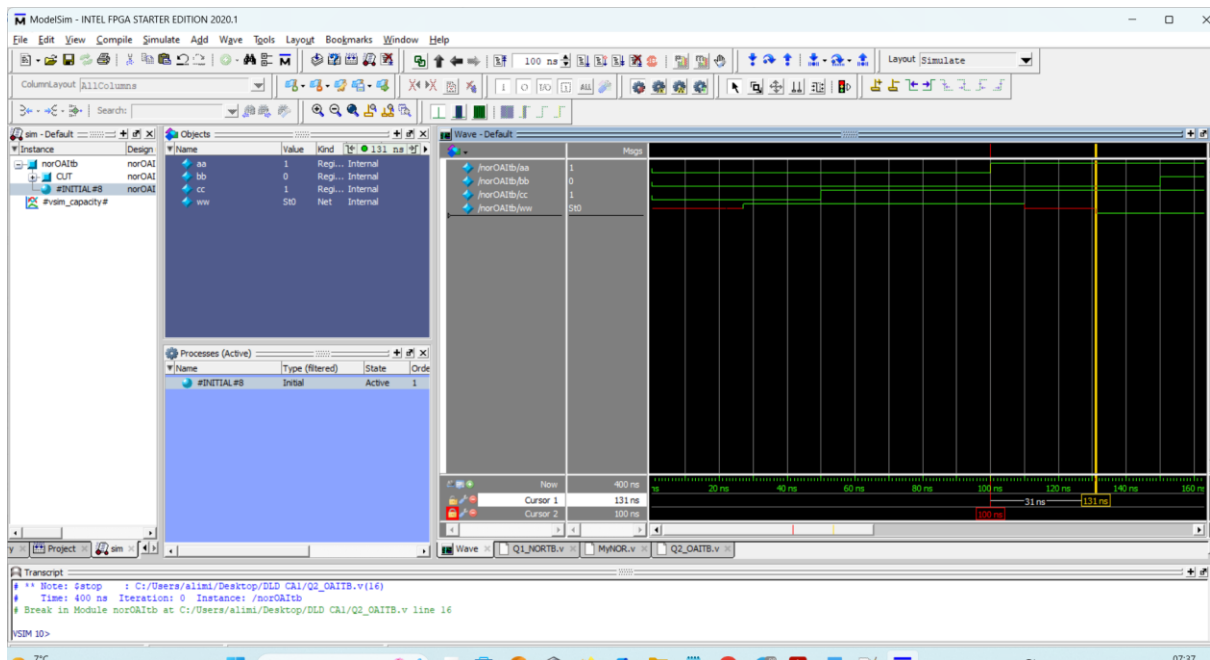
Verilog testbench

```
*C:\Program Files\Notepad++\new 1.v - Notepad++ [Administrator]
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window 2
new 1.v
1 timescale 1ns/1ns
2 module norOAITb();
3 reg aa=0;
4 reg bb=0;
5 reg cc=0;
6 wire ww;
7 norOAI CUT (aa,bb,cc,ww);
8 initial begin
9 #50 cc = 1;
10 #50 aa = 1;
11 #50 bb = 1;
12 #50 cc = 0;
13 #50 cc = 1;
14 #50 bb = 0;
15 #50 aa = 0;
16 #50 $stop;
17 end
18 endmodule
```

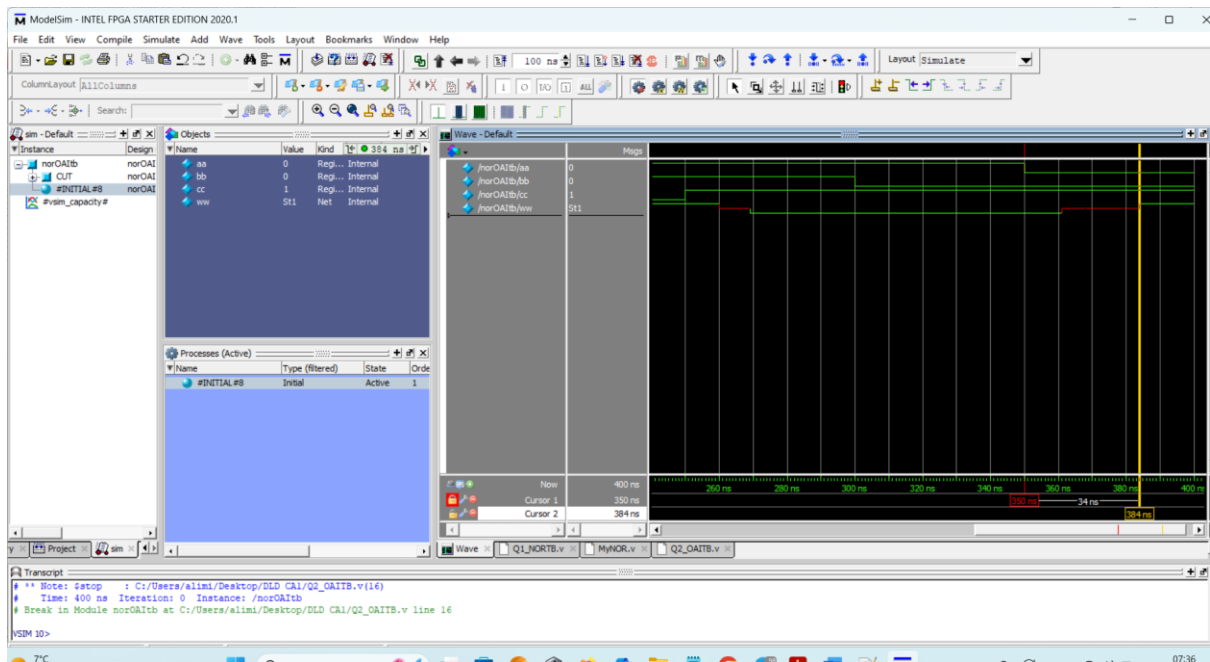
Verilog file length: 256 lines: 18 Ln: 18 Col: 10 Pos: 257 Windows (CR LF) UTF-8 INS



worst case to 0



worst case to 1

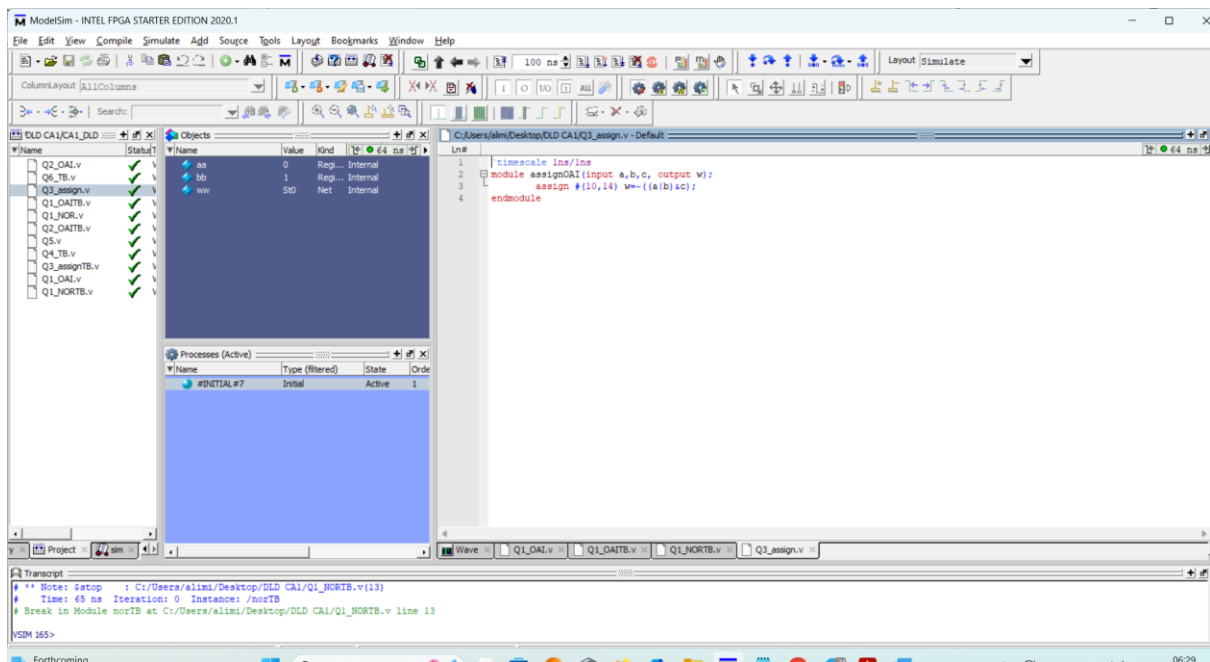


3)

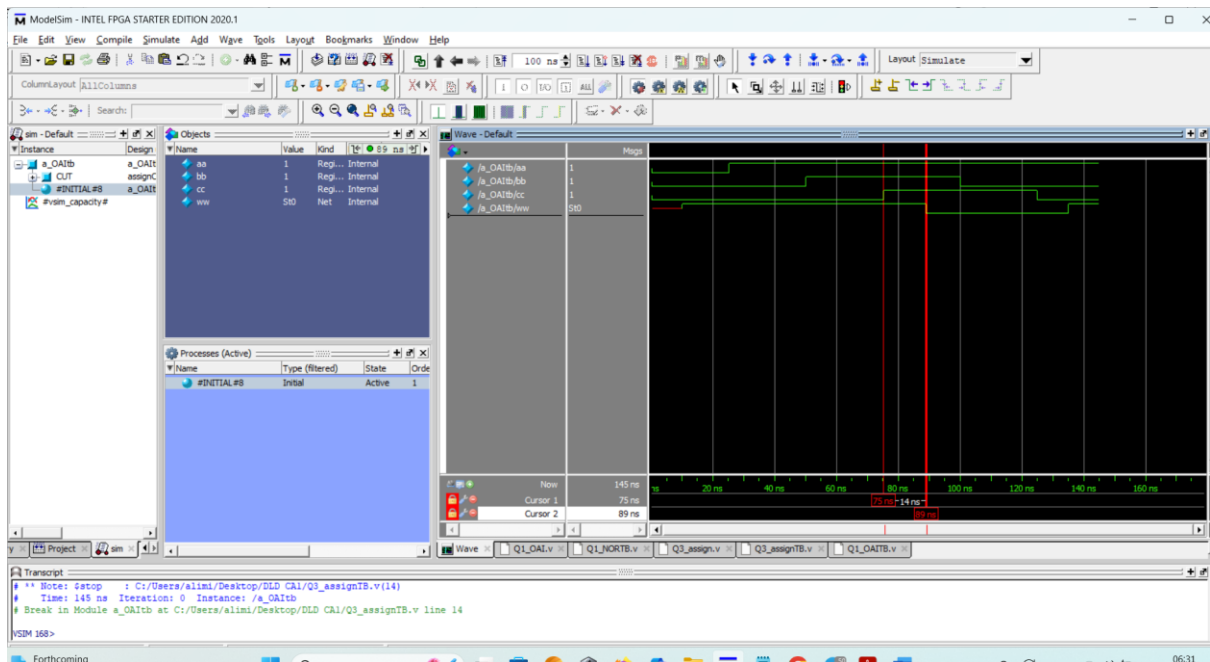
Verilog description

```
*C:\Program Files\Notepad++\new 1.v - Notepad++ [Administrator]
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window 2
new 1.v
1 timescale 1ns/1ns
2 module assignOAI(input a,b,c, output w);
3   assign # (10,14) w=~((a|b)&c);
4 endmodule

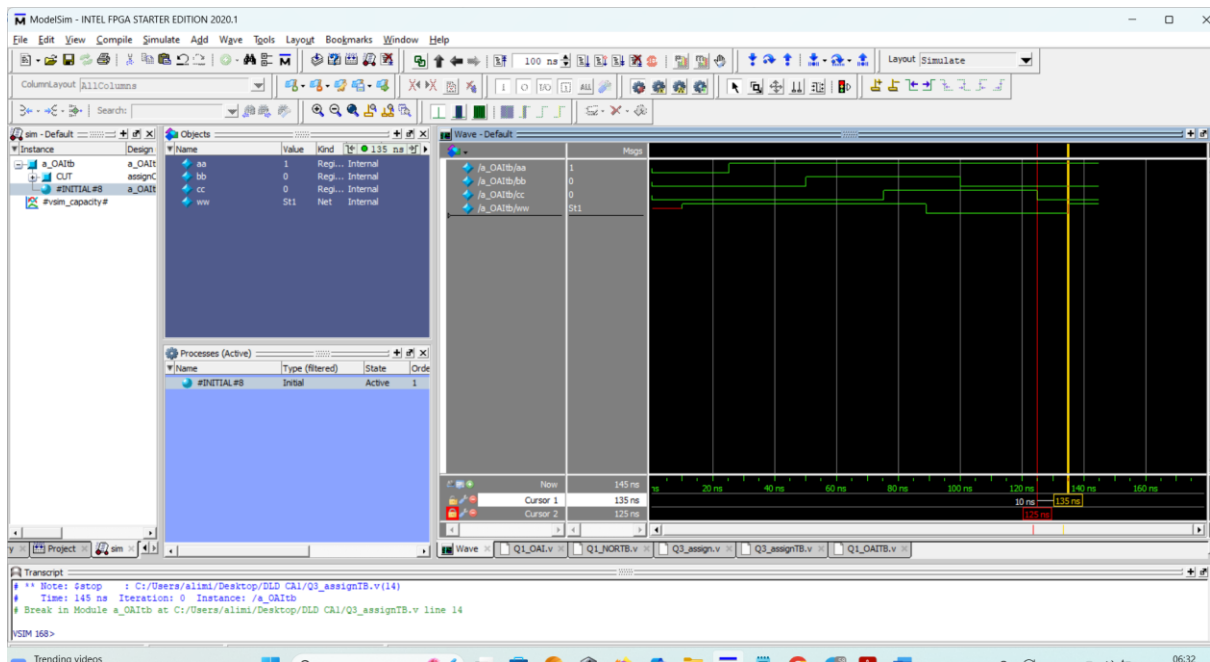
Verilog file length: 103 lines: 4 Ln: 4 Col: 10 Pos: 104 Windows (CR LF) UTF-8 INS
```



Verilog testbench



Worst case to 1:



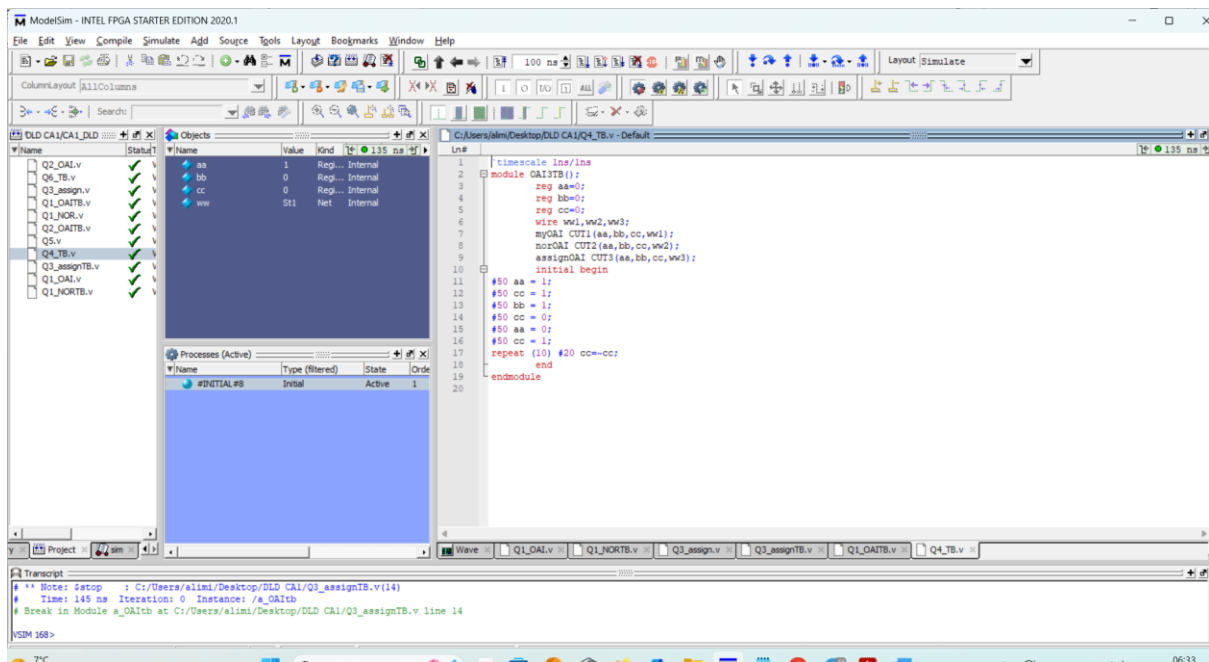
4)

Verilog testbench

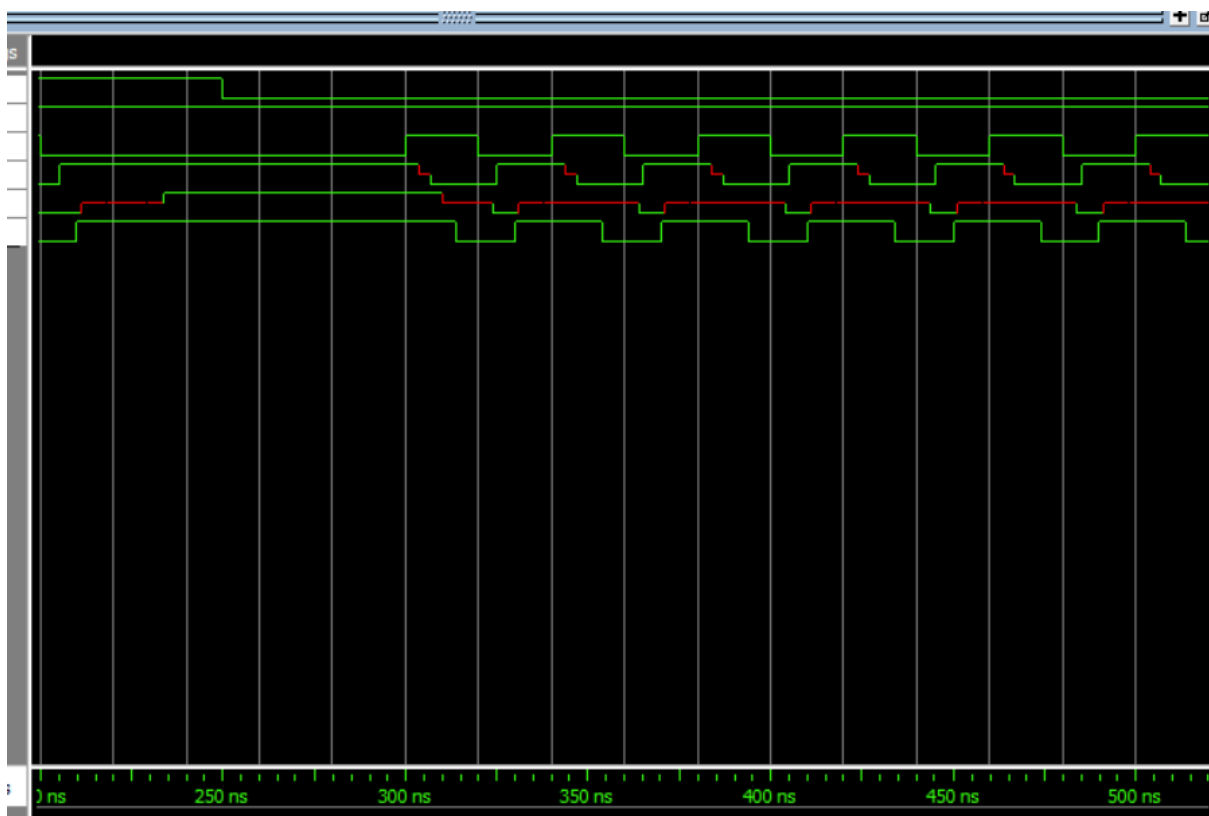
```

1 timescale 1ns/1ns
2 module QAI3TB1();
3     reg aa=0;
4     reg bb=0;
5     reg cc=0;
6     wire ww1,ww2,ww3;
7     myOAI CUT1(aa,bb,cc,ww1);
8     norOAI CUT2(aa,bb,cc,ww2);
9     assignOAI CUT3(aa,bb,cc,ww3);
10    initial begin
11        #50 aa = 1;
12        #50 cc = 1;
13        #50 bb = 1;
14        #50 cc = 0;
15        #50 aa = 0;
16        #50 cc = 1;
17        repeat (10) #20 cc==cc;
18    end
19 endmodule

```



Wave form:



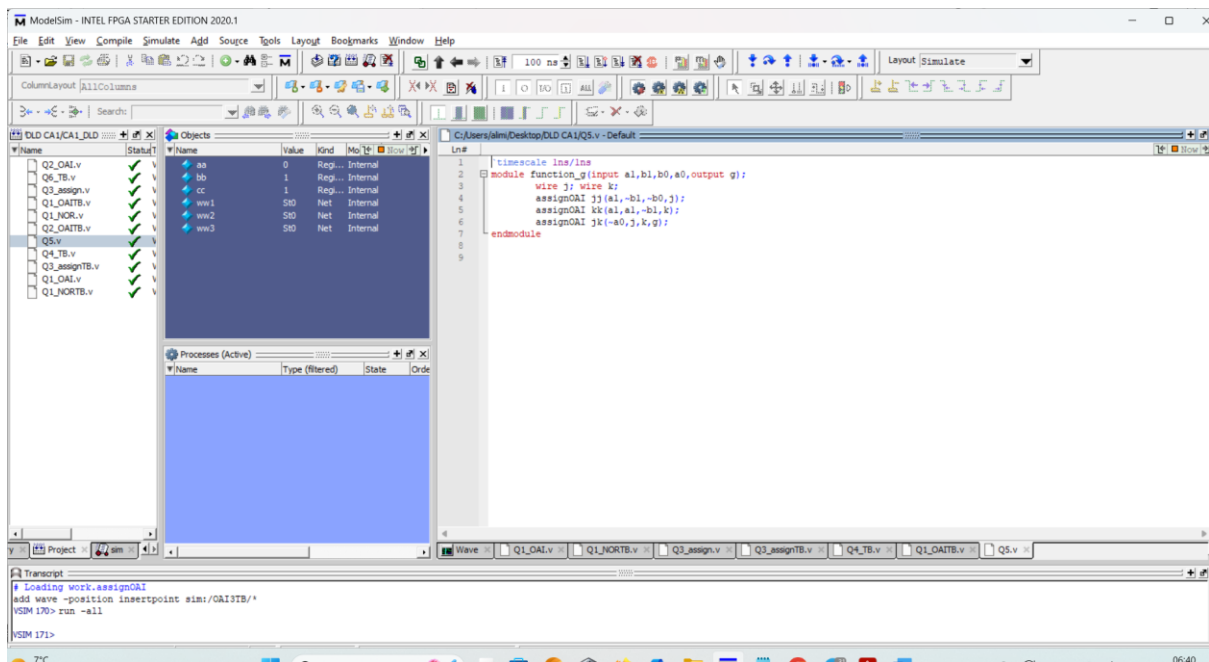
It seems that the second design (using NOR) is not suitable for us because it frequently produces an **x** output. The third design (using assign) performs better since it never results in **x** or **z**, but it has a longer delay compared to the first design. Ultimately, the first design is the best choice because it has the least amount of delay among the three options.

Q5)

Verilog description


```
*C:\Program Files\Notepad++\new 1.v - Notepad++ [Administrator]
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window 2
new 1.v
1 timescale 1ns/1ns
2 module function g(input a1,b1,b0,a0,output g):
3     wire j; wire k;
4     assignOAI jj(a1,-b1,-b0,j);
5     assignOAI kk(a1,a1,-b1,k);
6     assignOAI jk(a0,j,k,g);
7 endmodule
8
```

Verilog file length: 184 lines: 8 Ln: 8 Col: 5 Pos: 185 Windows (CR LF) UTF-8 INS



Q6)

Verilog TEST BENCH

C:\Program Files\Notepad++\new 1.v - Notepad++ [Administrator]

File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?

new 1.v

```
1 timescale 1ns/1ns
2 module gTB();
3   reg aa1=0, aa0=0, bb1=0, bb0=0;
4   wire gg;
5   function g CUT(aa1,aa0,bb1,bb0,gg);
6   initial begin
7     #5 aa0=1;
8     #25 bb0=1;
9     #25 aa1=1;
10    #25 aa1=0;
11    #25 bb1=1;
12    #25 $stop;
13  end
14 endmodule
```

Verilog file length: 226 lines: 14 Ln: 14 Col: 10 Pos: 227 Windows (CR LF) UTF-8 INS

ModelSim - INTEL FPGA STARTER EDITION 2020.1

File Edit View Compile Simulate Add Source Tools Layout Bookmarks Window Help

ColumnsLayout AllColumns

Q2_OAL.v Q6_TB.v Q3_assign.v Q1_OALTB.v Q1_NORT.v Q2_OALTB.v Q5.v Q4_TB.v Q3_assignTB.v Q1_OAL.v Q1_NORTTB.v

Name	Value	Kind	Mo	Th
aa	0	Reg...	Internal	
bb	1	Reg...	Internal	
cc	1	Reg...	Internal	
ww1	S0	Net	Internal	
ww2	S0	Net	Internal	
ww3	S0	Net	Internal	

Processes (Active)

Name	Type (Filtered)	State	Order
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C:\Users\adm\Desktop\OLD CA1\Q6_TB.v - Default

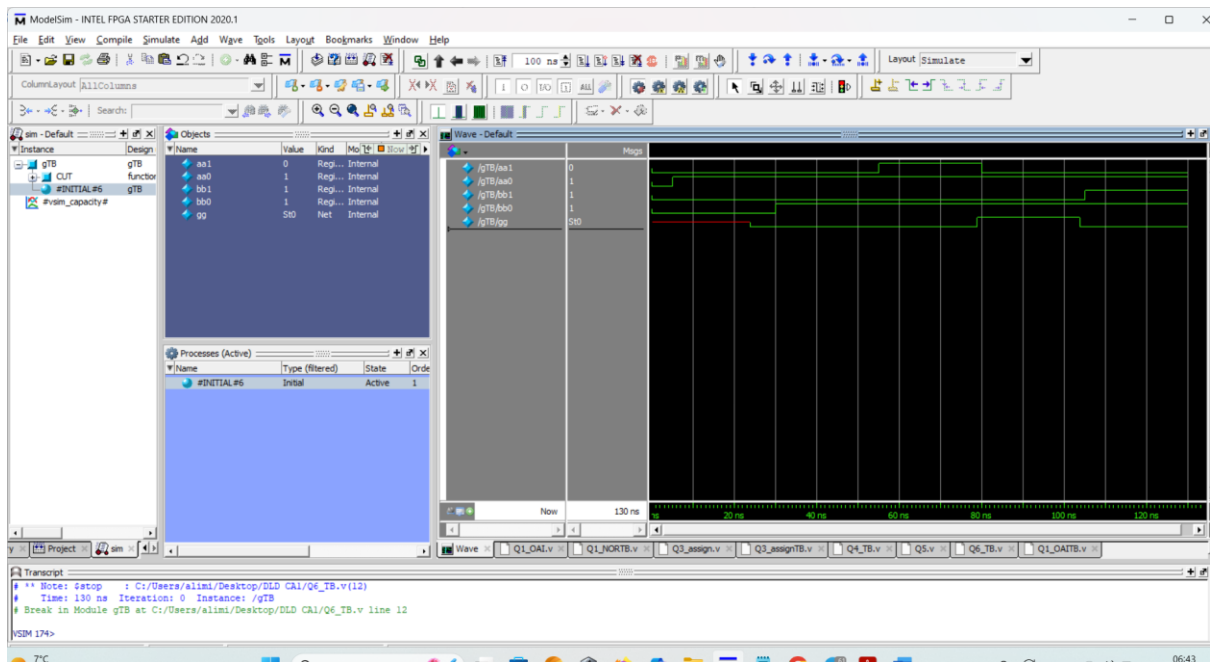
```
1 timescale 1ns/1ns
2 module gTB();
3   reg aa1=0, aa0=0, bb1=0, bb0=0;
4   wire gg;
5   function g CUT(aa1,aa0,bb1,bb0,gg);
6   initial begin
7     #5 aa0=1;
8     #25 bb0=1;
9     #25 aa1=1;
10    #25 aa1=0;
11    #25 bb1=1;
12    #25 $stop;
13  end
14 endmodule
```

Wave Q1_OAL.v Q1_NORTTB.v Q3_assign.v Q3_assignTB.v Q4_TB.v Q1_OALTB.v Q5.v Q6_TB.v

Transcript

```
# Loading work.assignOAL
add wave -position insertpoint sim:/QAL9TB/*
V$M170> run -all
V$M171>
```

7°C 06:41



a_1	a_0	b_1	b_0	^{decim} A	^{decimal} B	g
0	0	0	0	0	0	0
0	0	0	1	0	1	0
0	0	1	0	0	2	0
0	0	1	1	0	3	0
0	1	0	0	1	0	1
0	1	0	1	1	1	0
0	1	1	0	1	2	0
0	1	1	1	1	3	0
1	0	0	0	9	2	0
1	0	0	1	2	1	1
1	0	1	0	2	2	0
1	0	1	1	2	3	0
1	1	0	0	3	0	1
1	1	0	1	3	1	1
1	1	1	0	3	2	1
1	1	1	1	3	3	0

whenever A is greater than B

g is 1, O.W is 0.

↓
output

As you see, the function of circuit is : IF $A > B$: g is 1

O.W: g is 0

We also see the predicted results in TEST BENCH.